

From Thermodynamic Neurons to Floquet-Buffered Non-Markovian Logic Gates

Implementation and Theory Bridge

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Abstract

This note explains how the current codebase connects to the proposed research idea. The baseline thermodynamic-neuron paper is first reconstructed in the Markovian reset model. The extension then asks what changes when the bath is structured, has memory, and is not directly attached to the logical qubit. The proposed answer is to insert a periodically driven Floquet buffer between the logical subsystem and the environment. The buffer acts as a dynamical filter and isolation layer. Since direct simulation of a many-qubit system plus structured reservoirs grows exponentially, the environmental memory is represented as a tensor network, specifically a process tensor or PT-MPO. The operational output is not an abstract current alone, but a measurable logical distinguishability of output states.

1 What Has Already Been Reconstructed

The first completed part of the project is the Markovian baseline thermodynamic neuron. This corresponds to the weak-coupling, reset-model picture of the baseline paper. In this model, logical inputs are encoded as inverse temperatures of thermal reservoirs, and the output is encoded in the final inverse temperature of a finite reservoir.

The basic thermal occupation of a qubit with energy gap ϵ is

$$g(\beta\epsilon) = \frac{1}{1 + e^{\beta\epsilon}}. \quad (1)$$

For the three-qubit collector of the NOT gate, the virtual inverse temperature is

$$\beta_v = \frac{\epsilon_0}{\epsilon_z}\beta_0 - \frac{\epsilon_1}{\epsilon_z}\beta_1, \quad \epsilon_z = \epsilon_0 - \epsilon_1. \quad (2)$$

The reset-model collector and modulator currents are

$$j_C = \mu\epsilon_z [g_z(\beta_z) - g_z(\beta_v)], \quad (3)$$

$$j_M = \mu'\epsilon_z [g_z(\beta_z) - g_z(\beta_r)]. \quad (4)$$

Here $g_z(\beta) = g(\beta\epsilon_z)$. The finite output reservoir evolves as

$$\dot{\beta}_z(t) = -\frac{\beta_z(t)^2}{C} [j_C + j_M]. \quad (5)$$

The code in `Scripts/baseline_paper_reconstruction/` implements this model and generates the baseline NOT, NOR, 3-majority, and XOR responses. This is the reference point: every extension must be compared against it.

2 Why The Baseline Is Not The Final Theory

The baseline model is clean because the bath is memoryless. The system state $\rho_S(t)$ evolves using only its current value:

$$\dot{\rho}_S(t) = \mathcal{L}\rho_S(t). \quad (6)$$

This is a Markovian GKSL-type picture. It assumes weak system-bath coupling, rapid bath correlation decay, and no long-lived information returning from the reservoir.

For a structured bath, this is no longer safe. The bath may remember what the system did earlier. Then the reduced dynamics has a memory kernel:

$$\dot{\rho}_S(t) = -i[H_S, \rho_S(t)] + \int_0^t \mathcal{K}(t - \tau) \rho_S(\tau) d\tau. \quad (7)$$

This is the Nakajima–Zwanzig form. It says, in plain language, that the present motion of the system depends on its past. The object $\mathcal{K}(t - \tau)$ is the memory kernel. When the bath memory is very short, the kernel becomes sharply peaked and one recovers the ordinary Markovian equation. When the memory is long, the Markovian approximation may miss revivals, backflow, and modified relaxation.

This is why the project cannot jump directly from the reset-model baseline to a strong claim about non-Markovian thermodynamic gates. First one must show that the non-Markovian solver is stable and that the chosen observable is meaningful.

3 The Floquet Buffer Idea

The proposed architecture inserts a driven intermediate system between the logical subsystem and the bath:

$$S \longleftrightarrow F(t) \longleftrightarrow B. \quad (8)$$

The direct architecture is

$$S \longleftrightarrow B. \quad (9)$$

Here S is the logical subsystem, $F(t)$ is the periodically driven Floquet buffer, and B is the bath. A minimal buffer Hamiltonian is

$$H_F(t) = \frac{\epsilon_F}{2} \sigma_z + A \cos(\Omega t) \sigma_x, \quad (10)$$

with period

$$T = \frac{2\pi}{\Omega}. \quad (11)$$

The logical qubit-buffer interaction can be chosen as

$$H_{SF} = g_{SF} \sigma_x^{(S)} \otimes \sigma_x^{(F)}. \quad (12)$$

The complete minimal bridge Hamiltonian is therefore

$$H(t) = H_S + H_F(t) + H_{SF} + H_B + H_{FB}. \quad (13)$$

The physical role of the buffer is simple:

- it prevents the logical qubit from touching the bath directly;
- it creates drive-controlled sidebands at frequencies shifted by multiples of Ω ;
- it can suppress unwanted transitions and enhance desired transitions;

- it gives a possible gain/isolation stage, analogous in spirit to a CMOS inverter.

In a weak Markovian treatment of the buffer-bath contact, one obtains a time-periodic effective generator

$$\dot{\rho}_S(t) = \mathcal{L}_{\text{eff}}(t)\rho_S(t), \quad \mathcal{L}_{\text{eff}}(t+T) = \mathcal{L}_{\text{eff}}(t). \quad (14)$$

The transition rates depend on bath correlations evaluated at shifted frequencies

$$\omega + n\Omega, \quad (15)$$

where $n \in \mathbb{Z}$. This is the origin of Floquet filtering.

4 What The Current Floquet Code Does

The code in `Scripts/floquet_buffer_extension/` implements the first bridge model:

$$S \longleftrightarrow F(t) \longleftrightarrow B. \quad (16)$$

It compares this against the direct baseline

$$S \longleftrightarrow B. \quad (17)$$

For two different initial logical states, $\rho_S^{(0)}(0)$ and $\rho_S^{(1)}(0)$, the code evolves the output states and computes the trace distance

$$D(t) = \frac{1}{2} \left\| \rho_S^{(0)}(t) - \rho_S^{(1)}(t) \right\|_1. \quad (18)$$

This is the operational distinguishability of the two logical outputs. If $D = 1$, the states are perfectly distinguishable. If $D = 0$, they are indistinguishable.

The code also estimates the work injected by the drive through

$$W_{\text{drive}} = \int_0^{t_f} \text{Tr} \left[\rho(t) \frac{\partial H_F(t)}{\partial t} \right] dt. \quad (19)$$

For

$$H_F(t) = \frac{\epsilon_F}{2} \sigma_z + A \cos(\Omega t) \sigma_x, \quad (20)$$

we have

$$\frac{\partial H_F(t)}{\partial t} = -A\Omega \sin(\Omega t) \sigma_x. \quad (21)$$

The first default run gives

$$D_{\text{direct}}(t_f) \approx 0.135, \quad (22)$$

$$D_{\text{buffer}}(t_f) \approx 0.182, \quad (23)$$

$$W_{\text{drive}} \approx 0.460. \quad (24)$$

Thus the first parameter set shows the intended trade-off:

$$\text{better output distinguishability} \quad \text{at the cost of positive drive work.} \quad (25)$$

5 Why Tensor Networks Are Needed

If one explicitly simulates the system plus bath, the Hilbert space grows exponentially. Suppose the system has dimension d_S and the bath is represented by N local sites of dimension d . The full state dimension is

$$d_S d^N. \quad (26)$$

This is impossible for large N .

For non-Markovian open systems, the difficulty can also be seen in time. A bath with memory couples the system at time t_k to its earlier states at times t_j . Naively, storing all multi-time correlations scales exponentially with the number of time steps.

The process tensor solves this by representing the environment's influence as a tensor network:

$$\Upsilon_{n:0} = \sum_{\{\alpha_k, \beta_k\}} \text{Tr} \left[B_{\alpha_{n-1}, \beta_{n-1}}^{[n-1]} \cdots B_{\alpha_0, \beta_0}^{[0]} \right] |\alpha_{n-1} \cdots \alpha_0\rangle \langle \beta_{n-1} \cdots \beta_0|. \quad (27)$$

This is a matrix product operator along the time direction. Its internal bond dimension χ measures how much memory must be carried from the past to the future.

If the bath is Markovian, then

$$\chi = 1. \quad (28)$$

If the bath is non-Markovian but compressible, then

$$1 < \chi \ll d^{2n}. \quad (29)$$

This is the key reason tensor networks prevent exponential blow-up: the full memory object is not stored exactly as an exponentially large tensor; it is compressed by singular value decomposition while keeping the physically important temporal correlations.

6 Single Example: From Nakajima–Zwanzig To Output Logic

Consider one logical qubit coupled to a structured bath through a Floquet buffer:

$$S \longleftrightarrow F(t) \longleftrightarrow B. \quad (30)$$

Step 1: Exact Reduced Dynamics

The full density matrix obeys

$$\dot{\rho}_{SFB}(t) = -i[H_{SFB}(t), \rho_{SFB}(t)]. \quad (31)$$

The reduced logical state is

$$\rho_S(t) = \text{Tr}_{FB} \rho_{SFB}(t). \quad (32)$$

Step 2: Memory Kernel View

Tracing out the buffer and bath gives a memory equation

$$\dot{\rho}_S(t) = -i[H_S, \rho_S(t)] + \int_0^t \mathcal{K}_F(t, \tau) \rho_S(\tau) d\tau. \quad (33)$$

The kernel now depends on the driven buffer. By tuning A and Ω , one changes $\mathcal{K}_F$ and therefore changes how information leaves or returns to the logical qubit.

Step 3: Tensor Network Representation

Instead of explicitly constructing $\mathcal{K}_F$, the process-tensor method builds the multi-time influence object $\Upsilon_{n:0}$. The simulation then contracts the logical system operations with this environment tensor network:

$$\rho_S(t_n) = \Upsilon_{n:0} \star (\mathcal{A}_{n-1} \otimes \cdots \otimes \mathcal{A}_0), \quad (34)$$

where $\star$ means tensor contraction and $\mathcal{A}_k$ are the system operations at each time step.

Step 4: Logical Output

For a binary gate, prepare two input states and evolve them:

$$\rho_S^{(0)}(0) \rightarrow \rho_S^{(0)}(t_f), \quad \rho_S^{(1)}(0) \rightarrow \rho_S^{(1)}(t_f). \quad (35)$$

The logical distinguishability is

$$D_{\text{out}} = \frac{1}{2} \left\| \rho_S^{(0)}(t_f) - \rho_S^{(1)}(t_f) \right\|_1. \quad (36)$$

The Floquet buffer is useful only if it improves D_{out} or improves robustness at an acceptable drive cost.

7 How The Codebase Now Maps To The Theory

Theory object	Code location	Purpose
Baseline reset model	baseline reconstruction folder	Markovian thermodynamic neuron
Structured bath benchmark	non-Markovian extension folder	OQuPy/PT-MPO calibration
Floquet buffer bridge	Floquet-buffer extension folder	Test driven isolation layer
Convergence scans	non-Markovian outputs folder	Verify numerical stability

8 Next Scientific Step

The next step is not to claim that the full thermodynamic gate has been solved. The next step is to connect the Floquet-buffer bridge to the non-Markovian machinery:

$$S \longleftrightarrow F(t) \longleftrightarrow B_{\text{structured}}, \quad (37)$$

with the structured bath represented by a process tensor or chain mapping. The first target should remain a single logical qubit. Only after this is stable should the full three-qubit NOT or NOR thermodynamic neuron be attempted.